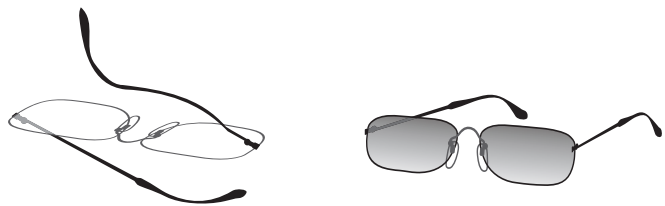


Examiner
only

5. (a) Smart materials are used to make the frames and lenses of certain spectacles.



Give the names of the different types of smart material used. Describe the unusual property of each. [2]

Frames

.....

.....

Lenses

.....

.....



Examiner
only

(b) Titanium dioxide has been used in sun creams for decades. Sun screens available today use nano-scale titanium dioxide particles. Some people believe using creams containing nano-particles is unsafe.

(i) Give the advantage of using nano-scale titanium dioxide particles rather than larger titanium dioxide particles to make sun screens. [1]

.....

.....

(ii) Explain why some people are concerned about the use of nano-scale titanium dioxide particles in sun screens. [2]

.....

.....

.....

(iii) **Approximately** how many times bigger are common titanium dioxide particles ($3 \times 10^{-7}\text{m}$) than nano-scale titanium dioxide particles ($2.5 \times 10^{-10}\text{m}$)? [2]

Answer =

7

3410UB01
11

